

```
import pandas as pd
```

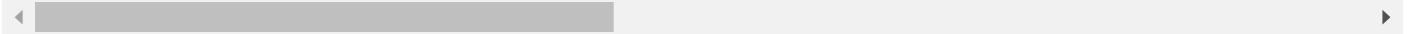
```
df = pd.read_csv('/content/drive/MyDrive/AI ML Bootcamp/Week 1/Invistico_Airline.csv')
```

```
df.head()
```



	satisfaction	Gender	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	...	Online support	Ease of Online booking	On-board service	se
0	satisfied	Female	Loyal Customer	65	Personal Travel	Eco	265	0		0	0	...	2	3	3
1	satisfied	Male	Loyal Customer	47	Personal Travel	Business	2464	0		0	0	...	2	3	4
2	satisfied	Female	Loyal Customer	15	Personal Travel	Eco	2138	0		0	0	...	2	2	3
3	satisfied	Female	Loyal Customer	60	Personal Travel	Eco	623	0		0	0	...	3	1	1
4	satisfied	Female	Loyal Customer	70	Personal Travel	Eco	354	0		0	0	...	4	2	2

5 rows × 23 columns



```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
df.isnull().sum()
```



	0
satisfaction	0
Gender	0
Customer Type	0
Age	0
Type of Travel	0
Class	0
Flight Distance	0
Seat comfort	0
Departure/Arrival time convenient	0
Food and drink	0
Gate location	0
Inflight wifi service	0
Inflight entertainment	0
Online support	0
Ease of Online booking	0
On-board service	0
Leg room service	0
Baggage handling	0
Checkin service	0
Cleanliness	0
Online boarding	0
Departure Delay in Minutes	0
Arrival Delay in Minutes	393

```
newdf = df.dropna()
```

```
newdf.isnull().sum()
```



	0
satisfaction	0
Gender	0
Customer Type	0
Age	0
Type of Travel	0
Class	0
Flight Distance	0
Seat comfort	0
Departure/Arrival time convenient	0
Food and drink	0
Gate location	0
Inflight wifi service	0
Inflight entertainment	0
Online support	0
Ease of Online booking	0
On-board service	0
Leg room service	0
Baggage handling	0
Checkin service	0
Cleanliness	0
Online boarding	0
Departure Delay in Minutes	0
Arrival Delay in Minutes	0

```
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import IterativeImputer
```

```
imputer = IterativeImputer()
```

```
from sklearn.preprocessing import LabelEncoder
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import IterativeImputer
```

```
label_encoders = {}
for column in df.select_dtypes(include=['object']).columns:
    label_encoders[column] = LabelEncoder()
    df[column] = label_encoders[column].fit_transform(df[column].astype(str))
```

```
imputer = IterativeImputer()
newdf1 = pd.DataFrame(imputer.fit_transform(df), columns=df.columns)
```

```
newdf1.isnull().sum()
```



	0
satisfaction	0
Gender	0
Customer Type	0
Age	0
Type of Travel	0
Class	0
Flight Distance	0
Seat comfort	0
Departure/Arrival time convenient	0
Food and drink	0
Gate location	0
Inflight wifi service	0
Inflight entertainment	0
Online support	0
Ease of Online booking	0
On-board service	0
Leg room service	0
Baggage handling	0
Checkin service	0
Cleanliness	0
Online boarding	0
Departure Delay in Minutes	0
Arrival Delay in Minutes	0

```
new_df1 = imputer.fit_transform(df)
```

```
new_df1 = pd.DataFrame(new_df1, columns=df.columns)
```

```
new_df1.isnull().sum()
```



	0
satisfaction	0